

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------|------------|------|-------------------|------------------------|------------|------------------|------------------------|------------|--------------|------------------------|------------|------------------|------------------------|------------|
| Aenderungsbeschreibung/Description of Revision<br>war kZ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                | Kommt vor/Frequency                                                                                                                                                                                                                                                                                                                                                                                                        |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Buchst./Rev.Ltr.<br>-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Aenderungs-Nr./Revision Notice No.<br>PR047866 |                                                                                                                                                                                                                                                                                                                                                                                                                            | Bearbeitungsstatus/Lifecycle<br>Released |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Changes from Var1 Ed 01.00 to Var1 Ed 01.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Description of StatusStartProcedure and UserInstruction added                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Corrections:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Reg 1205/1206 from HI to LO (P-FuelPrefilter)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| SCR alarms/measuring points renamed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| New measuring points added:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Reg 220 added (T-ExhaustSCROutlet2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Reg 295 added ( MMS Errorcodes)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| New limits added:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Reg 1221 Item added (LM AL RA Tank Level Empty)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Reg 1100 added (LM LOLO P-Raw Water)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| New alarms added:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| AL Permanent Injection Rail2<br>AL MMS Combined Red<br>AL MMS Combined Yel<br>AL RA Tank Level Empty<br>SD Lube Oil Level Switch<br>LOLO P-Raw Water<br>AL Gas Inclusion<br>SD P-Coolant Intercooler<br>LO P-Coolant Intercooler<br>LOLO P-Coolant Intercooler<br>SD P-Intake Air After FilterA<br>LO P-Intake Air After FilterA<br>LOLO P-Intake AirAfterFilterA<br>SD P-Intake Air After FilterB<br>LO P-Intake Air After FilterB<br>LOLO P-Intake AirAfterFilterB<br>SD T-Coolant Intercooler<br>HI T-Coolant Intercooler<br>HIHI T-Coolant Intercooler<br>AL Inj. Drift Limit 1 Cyl.A1<br>AL Inj. Drift Limit 1 Cyl.A2<br>AL Inj. Drift Limit 1 Cyl.A3<br>AL Inj. Drift Limit 1 Cyl.A4<br>AL Inj. Drift Limit 1 Cyl.A5<br>AL Inj. Drift Limit 1 Cyl.A6<br>AL Inj. Drift Limit 1 Cyl.A7<br>AL Inj. Drift Limit 1 Cyl.A8<br>AL Inj. Drift Limit 1 Cyl.A9<br>AL Inj. Drift Limit 1 Cyl.A10<br>AL Inj. Drift Limit 2 Cyl.A1<br>AL Inj. Drift Limit 2 Cyl.A2<br>AL Inj. Drift Limit 2 Cyl.A3<br>AL Inj. Drift Limit 2 Cyl.A4<br>AL Inj. Drift Limit 2 Cyl.A5<br>AL Inj. Drift Limit 2 Cyl.A6<br>AL Inj. Drift Limit 2 Cyl.A7<br>AL Inj. Drift Limit 2 Cyl.A8<br>AL Inj. Drift Limit 2 Cyl.A9 |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Allgemeintoleranzen/General Tolerances ISO 2768-mK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                | Masse/Mass kg    Format/Size<br><b>A4</b>                                                                                                                                                                                                                                                                                                                                                                                  |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Tolerierung/Tolerancing ISO 8015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                | Masse/Size ISO 14405                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Ausf. u.Lieferung Tech.<br>Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                | Oberflaechenangaben nach<br>MTN5033                                                                                                                                                                                                                                                                                                                                                                                        |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Oberflaechenschutz<br>Surface Protection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                | Surface Specification per MTN5033                                                                                                                                                                                                                                                                                                                                                                                          |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Verwendbar f.Type<br>Applicable to Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                | GLNG Release<br>10                                                                                                                                                                                                                                                                                                                                                                                                         |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Projekt-/Auftrags-Nr.<br>Project/Order No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                | Allgemeine Fertigungsvorschriften<br>nach MMN332                                                                                                                                                                                                                                                                                                                                                                           |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Ref.-Nr./ Ref. No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                | Production Specification per<br>MMN 332                                                                                                                                                                                                                                                                                                                                                                                    |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Type1 Var01 ED01_00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                | Emission ID                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
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| WORD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Datum/Date                                     | Name                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Erstell.<br>Drawn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2018-09-10<br>12:20:09                         | schaeppers                                                                                                                                                                                                                                                                                                                                                                                                                 |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Bearb.<br>Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2019-09-11<br>09:44:43                         | schaeppers                                                                                                                                                                                                                                                                                                                                                                                                                 |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Norm<br>Std.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2019-09-11<br>09:44:43                         | ihmor stef                                                                                                                                                                                                                                                                                                                                                                                                                 |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Gepr.<br>Checked                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2019-09-11<br>09:44:43                         | ihmor stef                                                                                                                                                                                                                                                                                                                                                                                                                 |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Benennung/ Title<br><b>DOKUMENTATION</b><br>GLNG Fremdschnittstelle    Genoline  <br>Modbus TCP    NewGeneration<br><b>DOCUMENTATION</b><br>GLNG Foreign MCS Modbus<br>TCP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                | XZ00E52000016                                                                                                                                                                                                                                                                                                                                                                                                              |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| <br>MTU Friedrichshafen GmbH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                | Zeichnungs-Nr./Drawing No. <b>ZNG00019138</b>                                                                                                                                                                                                                                                                                                                                                                              |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |
| Beschreibung/Description<br>Type1 Var01 ED01_02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                | Blatt/ Sheet<br><b>1</b><br>von/of<br><b>27</b>                                                                                                                                                                                                                                                                                                                                                                            |                                          |      |            |      |                   |                        |            |                  |                        |            |              |                        |            |                  |                        |            |

T-468-0107

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| Aenderungsbeschreibung/Description of Revision<br>war kZ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                | Kommt vor/Frequency                                                                                                        |
| Buchst./Rev.Ltr.<br>-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Aenderungs-Nr./Revision Notice No.<br>PR047866 | Bearbeitungsstatus/Lifecycle<br>Released                                                                                   |
| <div>AL Inj. Drift Limit 2 Cyl.A10<br/>AL Inj. Drift Limit 1 Cyl.B1<br/>AL Inj. Drift Limit 1 Cyl.B2<br/>AL Inj. Drift Limit 1 Cyl.B3<br/>AL Inj. Drift Limit 1 Cyl.B4<br/>AL Inj. Drift Limit 1 Cyl.B5<br/>AL Inj. Drift Limit 1 Cyl.B6<br/>AL Inj. Drift Limit 1 Cyl.B7<br/>AL Inj. Drift Limit 1 Cyl.B8<br/>AL Inj. Drift Limit 1 Cyl.B9<br/>AL Inj. Drift Limit 1 Cyl.B10<br/>AL Inj. Drift Limit 2 Cyl.B1<br/>AL Inj. Drift Limit 2 Cyl.B2<br/>AL Inj. Drift Limit 2 Cyl.B3<br/>AL Inj. Drift Limit 2 Cyl.B4<br/>AL Inj. Drift Limit 2 Cyl.B5<br/>AL Inj. Drift Limit 2 Cyl.B6<br/>AL Inj. Drift Limit 2 Cyl.B7<br/>AL Inj. Drift Limit 2 Cyl.B8<br/>AL Inj. Drift Limit 2 Cyl.B9<br/>AL Inj. Drift Limit 2 Cyl.B10<br/>HI ETC Idle Speed<br/>LO RA Concentration<br/>LOLO RA Concentration<br/>SD RA Concentration<br/>HI RA Concentration<br/>HIHI RA Concentration<br/>LO RA Quality<br/>HI P-Exhaust Diff SCR2<br/>HIHI P-Exhaust Diff SCR2<br/>SD T-Exhaust SCR Outlet2<br/>HI T-Exhaust SCR Outlet2<br/>HIHI T-Exhaust SCR Outlet2<br/>AL PFM Leak Check Failed<br/>HI T-Reducing Agent SUDU F1<br/>HI T-Reducing Agent SUDU F2<br/>AL Hydraulics Fault SUDU1<br/>AL Hydraulics Fault SUDU2<br/>AL Hydraulics Fault SUDU3<br/>AL Hydraulics Fault SUDU4<br/>AL Hydraulics Fault SUDU5<br/>AL Hydraulics Fault SUDU6<br/>AL SD-Card Unaccessible<br/>SD Eng.Speed Crankshaft SaSy<br/>TD Engine Speed Crankshaft<br/>AL Coolant Leakage LT<br/>SD Coolant Leakage LT<br/>AL Oil Refill Tank Empty<br/>SD Oil Refill Tank Empty<br/>AL Com. Lost NOx EGAT Outlet2<br/>SD NOx EGAT Outlet 2<br/>AL Perm. Injection Detection</div> |                                                |                                                                                                                            |
| <div><br/>MTU Friedrichshafen GmbH</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                | <div>Zeichnungs-Nr./Drawing No.<br/><b>ZNG00019138</b></div> <div>Blatt/ Sheet<br/><b>2</b><br/>von/of<br/><b>27</b></div> |
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war kZ

Buchst./Rev.Ltr.  
-14Aenderungs-Nr./Revision Notice No.  
PR047866Bearbeitungsstatus/Lifecycle  
Released

## MCS Analog Measuring Points

|                                          |                           |
|------------------------------------------|---------------------------|
| Function Code                            | 03 Holding Register       |
|                                          | 16 Write Holding Register |
| Minimum Cycle Time                       | 200ms                     |
| Missing Data:                            | 0x7FFF                    |
| Sensor Defect                            | 0x8001                    |
| Sensor Defect Combined Registers (32Bit) | 0x80000001                |

| Register | Denomination                   | Min  | Max     | Unit  | Scaling | Read/Write |
|----------|--------------------------------|------|---------|-------|---------|------------|
| 0        | Engine Speed                   | 0    | 3000    | rpm   | 10      | R          |
| 1        | Engine Speed SaSy              | 0    | 3000    | rpm   | 10      | R          |
| 2        | Engine Speed Camshaft          | 0    | 3000    | rpm   | 10      | R          |
| 3        | Engine Speed Crankshaft        | 0    | 3000    | rpm   | 10      | R          |
| 4        | Idle Speed                     | 0    | 3000    | rpm   | 10      | R          |
| 5        | Nominal Speed                  | 0    | 3000    | rpm   | 10      | R          |
| 6        | Feedback Speed Demand          | 0    | 3000    | rpm   | 10      | R          |
| 7        | Feedback Speed Demand Eff.     | 0    | 3000    | rpm   | 10      | R          |
| 11       | ETC 1 Speed                    | 0    | 100     | krpm  | 100     | R          |
| 12       | ETC 2 Speed                    | 0    | 100     | krpm  | 100     | R          |
| 13       | ETC 3 Speed                    | 0    | 100     | krpm  | 100     | R          |
| 14       | ETC 4 Speed                    | 0    | 100     | krpm  | 100     | R          |
| 24       | Power Supply EIM/Starter Bat.  | 0    | 50      | V     | 100     | R          |
| 25       | ECU Power Supply Voltage       | 0    | 35      | V     | 100     | R          |
| 26       | U-SPU Main Power Supply        | 0    | 50      | V     | 100     | R          |
| 27       | U-SPU Emerg. Power Supply      | 0    | 50      | V     | 100     | R          |
| 30       | P-Lube Oil aft. Filter (ECU)   | 0    | 10      | bar   | 100     | R          |
| 31       | P-Lube Oil aft. Filter SaSy    | 0    | 10      | bar   | 100     | R          |
| 32       | P-Lube Oil A-Side ETCs         | 0    | 120     | bar   | 100     | R          |
| 33       | P-Lube Oil B-Side ETCs         | 0    | 120     | bar   | 100     | R          |
| 34       | P-Oil after Level Pump         | 0    | 10      | bar   | 100     | R          |
| 35       | P-Lube Oil last bearing        | 0    | 10      | bar   | 100     | R          |
| 36       | P-Piston Cooling Oil           | 0    | 6       | bar   | 100     | R          |
| 37       | P-Piston Cool.Oil Filter Diff. | 0    | 10      | bar   | 100     | R          |
| 38       | P-Lube Oil before Filter (ECU) | 0    | 10      | bar   | 100     | R          |
| 39       | P-Oil Filter Difference        | 0    | 16      | bar   | 100     | R          |
| 42       | P-Oil Refill Pump              | 0    | 6       | bar   | 100     | R          |
| 43       | P-Coolant                      | 0    | 6       | bar   | 100     | R          |
| 44       | P-Raw Water before Pump        | 0    | 6       | bar   | 100     | R          |
| 45       | P-Raw Water                    | 0    | 6       | bar   | 100     | R          |
| 46       | P-Fuel                         | 0    | 15      | bar   | 100     | R          |
| 47       | P-Fuel before Filter (ECU)     | 0    | 10      | bar   | 100     | R          |
| 48       | P-Fuel Prefilter Diff.         | 0    | 10      | bar   | 100     | R          |
| 49       | P-Fuel Filter Difference       | 0    | 16      | bar   | 100     | R          |
| 50       | P-Fuel Common Rail A           | 0    | 2500,00 | bar   | 10      | R          |
| 51       | P-Fuel Common Rail B           | 0    | 2500,00 | bar   | 10      | R          |
| 52       | P-Fuel Common Rail A SaSy      | 0    | 2500,00 | bar   | 10      | R          |
| 53       | P-Fuel Common Rail B SaSy      | 0    | 2500,00 | bar   | 10      | R          |
| 54       | P-Crankcase                    | -70  | 70      | mbar  | 100     | R          |
| 55       | P-Intake Air B                 | -70  | 70      | mbar  | 100     | R          |
| 56       | P-Charge Air A                 | 0    | 6       | bar   | 100     | R          |
| 57       | P-Charge Air B                 | 0    | 6       | bar   | 100     | R          |
| 58       | P-Control Air                  | 0    | 10      | bar   | 100     | R          |
| 59       | P-Start Air                    | 0    | 50      | bar   | 100     | R          |
| 60       | P-Exhaust back                 | 0    | 200     | bar   | 100     | R          |
| 62       | UTC Seconds                    | 0    | 59      | digit | 1       | R          |
| 63       | UTC Minutes                    | 0    | 59      | digit | 1       | R          |
| 64       | UTC Hour                       | 0    | 23      | digit | 1       | R          |
| 65       | Month                          | 1    | 12      | digit | 1       | R          |
| 66       | Day                            | 1    | 31      | digit | 1       | R          |
| 67       | Year                           | 1970 | 2060    | digit | 1       | R          |
| 70       | T-Lube Oil                     | -20  | 120     | degC  | 10      | R          |
| 71       | T-Lube Oil ETC                 | -20  | 120     | degC  | 10      | R          |
| 75       | T-Fuel bef.Low Press Pump      | -20  | 120     | degC  | 10      | R          |
| 76       | T-Fuel Common Rail A           | -20  | 120     | degC  | 10      | R          |
| 77       | T-Fuel Common Rail B           | -20  | 120     | degC  | 10      | R          |
| 78       | T-Coolant (ECU)                | -20  | 120     | degC  | 10      | R          |
| 79       | T-Coolant SaSy                 | -20  | 120     | degC  | 10      | R          |
| 80       | T-Raw Water                    | -20  | 120     | degC  | 10      | R          |
| 81       | T-Intake Air                   | -20  | 120     | degC  | 10      | R          |



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| Register | Denomination                      | Min         | Max        | Unit  | Scaling | Read/Write |
|----------|-----------------------------------|-------------|------------|-------|---------|------------|
| 82       | T-Charge Air A                    | -20         | 120        | degC  | 10      | R          |
| 83       | T-Charge Air B                    | -20         | 120        | degC  | 10      | R          |
| 84       | T-Charge Air Seq Ctrl Valve       | 0           | 1000       | degC  | 10      | R          |
| 85       | T-Exhaust Combined A              | -20         | 800        | degC  | 10      | R          |
| 86       | T-Exhaust before ETC (Combined B) | -20         | 800        | degC  | 10      | R          |
| 87       | T-Exhaust Combined C              | -20         | 1000       | degC  | 10      | R          |
| 88       | T-Exhaust Combined D              | -20         | 1000       | degC  | 10      | R          |
| 89       | T-Exhaust A1                      | 0           | 1000       | degC  | 10      | R          |
| 90       | T-Exhaust A2                      | 0           | 1000       | degC  | 10      | R          |
| 91       | T-Exhaust A3                      | 0           | 1000       | degC  | 10      | R          |
| 92       | T-Exhaust A4                      | 0           | 1000       | degC  | 10      | R          |
| 93       | T-Exhaust A5                      | 0           | 1000       | degC  | 10      | R          |
| 94       | T-Exhaust A6                      | 0           | 1000       | degC  | 10      | R          |
| 95       | T-Exhaust A7                      | 0           | 1000       | degC  | 10      | R          |
| 96       | T-Exhaust A8                      | 0           | 1000       | degC  | 10      | R          |
| 97       | T-Exhaust A9                      | 0           | 1000       | degC  | 10      | R          |
| 98       | T-Exhaust A10                     | 0           | 1000       | degC  | 10      | R          |
| 99       | T-Exhaust B1                      | 0           | 1000       | degC  | 10      | R          |
| 100      | T-Exhaust B2                      | 0           | 1000       | degC  | 10      | R          |
| 101      | T-Exhaust B3                      | 0           | 1000       | degC  | 10      | R          |
| 102      | T-Exhaust B4                      | 0           | 1000       | degC  | 10      | R          |
| 103      | T-Exhaust B5                      | 0           | 1000       | degC  | 10      | R          |
| 104      | T-Exhaust B6                      | 0           | 1000       | degC  | 10      | R          |
| 105      | T-Exhaust B7                      | 0           | 1000       | degC  | 10      | R          |
| 106      | T-Exhaust B8                      | 0           | 1000       | degC  | 10      | R          |
| 107      | T-Exhaust B9                      | 0           | 1000       | degC  | 10      | R          |
| 108      | T-Exhaust B10                     | 0           | 1000       | degC  | 10      | R          |
| 109      | T-Exhaust Mean                    | 0           | 1000       | degC  | 10      | R          |
| 110      | T-Bearing 1                       | 0           | 200        | degC  | 10      | R          |
| 111      | T-Bearing 2                       | 0           | 200        | degC  | 10      | R          |
| 112      | T-Bearing 3                       | 0           | 200        | degC  | 10      | R          |
| 113      | T-Bearing 4                       | 0           | 200        | degC  | 10      | R          |
| 114      | T-Bearing 5                       | 0           | 200        | degC  | 10      | R          |
| 115      | T-Bearing 6                       | 0           | 200        | degC  | 10      | R          |
| 116      | T-Bearing 7                       | 0           | 200        | degC  | 10      | R          |
| 117      | T-Bearing 8                       | 0           | 200        | degC  | 10      | R          |
| 118      | T-Bearing 9                       | 0           | 200        | degC  | 10      | R          |
| 119      | T-Bearing 10                      | 0           | 200        | degC  | 10      | R          |
| 120      | T-Bearing 11                      | 0           | 200        | degC  | 10      | R          |
| 121      | T-Bearing Mean                    | 0           | 200        | degC  | 10      | R          |
| 122      | T-Splash Oil B1                   | -40         | 150        | degC  | 10      | R          |
| 123      | T-Splash Oil B2                   | -40         | 150        | degC  | 10      | R          |
| 124      | T-Splash Oil B3                   | -40         | 150        | degC  | 10      | R          |
| 125      | T-Splash Oil B4                   | -40         | 150        | degC  | 10      | R          |
| 126      | T-Splash Oil B5                   | -40         | 150        | degC  | 10      | R          |
| 127      | T-Splash Oil B6                   | -40         | 150        | degC  | 10      | R          |
| 128      | T-Splash Oil B7                   | -40         | 150        | degC  | 10      | R          |
| 129      | T-Splash Oil B8                   | -40         | 150        | degC  | 10      | R          |
| 130      | T-Splash Oil B9                   | -40         | 150        | degC  | 10      | R          |
| 131      | T-Splash Oil B10                  | -40         | 150        | degC  | 10      | R          |
| 132      | T-Splash Oil Mean                 | -40         | 150        | degC  | 10      | R          |
| 133      | T-ECU                             | 0           | 100        | degC  | 10      | R          |
| 135      | T-Exhaust after Engine            | 0           | 400        | degC  | 10      | R          |
| 140      | ECU Failure Codes                 | 0           | 5000       | digit | 1       | R          |
| 141      | Torque Limitation Code            | -2147483646 | 2147483646 | digit | 1       | R          |
| 142      | SPU Interface Version HI          | -2147483646 | 2147483646 | digit | 1       | R          |
| 143      | SPU Interface Version LO          | -2147483646 | 2147483646 | digit | 1       | R          |
| 144      | MCS Version HI                    | -2147483646 | 2147483646 | digit | 1       | R          |
| 145      | MCS Version LO                    | -2147483646 | 2147483646 | digit | 1       | R          |
| 146      | SPU Alarm Groups coded static HI  | -2147483646 | 2147483646 | digit | 1       | R          |
| 147      | SPU Alarm Groups coded static LO  | -2147483646 | 2147483646 | digit | 1       | R          |
| 148      | SPU Alarm Groups coded dyn. HI    | -2147483646 | 2147483646 | digit | 1       | R          |
| 149      | SPU Alarm Groups coded dyn. LO    | -2147483646 | 2147483646 | digit | 1       | R          |
| 150      | Alarm Counter                     | 0           | 65535      | digit | 1       | R          |
| 152      | Status Start Procedure HI         | -2147483646 | 2147483646 | digit | 1       | R          |
| 153      | Status Start Procedure LO         | -2147483646 | 2147483646 | digit | 1       | R          |
| 155      | Trip Fuel Consumption HI          | 0           | 200000000  | l     | 1       | R          |
| 156      | Trip Fuel Consumption LO          | 0           | 200000000  | l     | 1       | R          |
| 157      | Total Fuel Consumption HI         | 0           | 200000000  | l     | 1       | R          |
| 158      | Total Fuel Consumption LO         | 0           | 200000000  | l     | 1       | R          |
| 159      | Actual Fuel Consumption           | 0           | 3000       | l/h   | 10      | R          |
| 160      | Mean Trip Fuel Consumption        | 0           | 3000       | l/h   | 10      | R          |



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| Register | Denomination                 | Min         | Max        | Unit  | Scaling | Read/Write |
|----------|------------------------------|-------------|------------|-------|---------|------------|
| 161      | Fuel Consumption Max         | 0           | 3000       | l/h   | 10      | R          |
| 162      | Fuel Consumption Act         | 0           | 100        | %     | 100     | R          |
| 163      | Fuel Consumption Mean Trip   | 0           | 100        | %     | 100     | R          |
| 165      | Injection Limitation 0-120%  | 0           | 120        | %     | 100     | R          |
| 166      | Injection in Relation to DBR | 0           | 120        | %     | 100     | R          |
| 170      | ECU Operat.Hours HI          | 0           | 100000     | h     | 1       | R          |
| 171      | ECU Operat.Hours LO          |             |            |       |         |            |
| 172      | Trip Duration HI             |             |            |       |         |            |
| 173      | Trip Duration LO             | 0           | 100000     | h     | 1       | R          |
| 176      | Engine Load Reserve          | 0           | 120        | %     | 100     | R          |
| 178      | Injection Quantity 0-120     | 0           | 120        | %     | 100     | R          |
| 183      | User Instruction             | 0           | 65535      | digit | 1       | R          |
| 204      | Cylinder Cutout Mask HI      | -2147483646 | 2147483646 | digit | 1       | R          |
| 205      | Cylinder Cutout Mask LO      |             |            |       |         |            |
| 206      | T-Exhaust SCR Inlet1         | 0           | 1000       | degC  | 10      | R          |
| 207      | T-Exhaust SCR Inlet2         | 0           | 1000       | degC  | 10      | R          |
| 208      | T-Exhaust SCR Outlet1        | 0           | 1000       | degC  | 10      | R          |
| 209      | P-Exhaust Diff SCR1          | 0           | 10         | bar   | 100     | R          |
| 210      | P-RA Filter Out At PFM       | 0           | 10         | bar   | 100     | R          |
| 211      | T-Reducing Agent At PFM      | 0           | 1000       | degC  | 10      | R          |
| 212      | RA Tank Level Norm           | 0           | 100        | %     | 100     | R          |
| 213      | P-Fuel Prefilter             | 0           | 10         | bar   | 100     | R          |
| 220      | T-Exhaust SCR Outlet 2       | 0           | 1000       | degC  | 10      | R          |
| 221      | Power Actual Gen             | 0           | 2000       | kW    | 10      | R          |
| 231      | T-Bearing GenDE              | -50         | 200        | degC  | 10      | R          |
| 232      | T-Bearing GenNDE             | -50         | 200        | degC  | 10      | R          |
| 233      | T-AirInlet Gen               | -50         | 200        | degC  | 10      | R          |
| 234      | T-AirOutlet Gen              | -50         | 200        | degC  | 10      | R          |
| 236      | T-WindingL1 Gen              | -50         | 200        | degC  | 10      | R          |
| 237      | T-WindingL2 Gen              | -50         | 200        | degC  | 10      | R          |
| 238      | T-WindingL3 Gen              | -50         | 200        | degC  | 10      | R          |
| 239      | T-WindingL4 Gen              | -50         | 200        | degC  | 10      | R          |
| 246      | T-Coolant Gen Out            | -50         | 200        | degC  | 10      | R          |
| 247      | T-Coolant Gen In             | -50         | 200        | degC  | 10      | R          |
| 295      | MMS Errorcodes               | 0           | 2147483646 | digit | 1       | R          |
| 500      | Adjust UTC Seconds           | 0           | 59         | digit | 1       | W          |
| 501      | Adjust UTC Minutes           | 0           | 59         | digit | 1       | W          |
| 502      | Adjust UTC Hours             | 0           | 23         | digit | 1       | W          |
| 503      | Adjust Day                   | 1           | 31         | digit | 1       | W          |
| 504      | Adjust Month                 | 1           | 12         | digit | 1       | W          |
| 505      | Adjust Year                  | 1975        | 2060       | digit | 1       | W          |
| 506      | Cmd Voltage Demand Gen       | 0           | 1000       | V     | 10      | W          |
| 507      | Cmd Current Demand Gen       | 0           | 3000       | A     | 10      | W          |
| 508      | Cmd Power Demand Gen         | 0           | 2000       | kW    | 10      | W          |



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## MCS Alarms

| Function Code                                                                                                                                                                        |         | 01 Read Coils        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------|
| Minimum Cycle Time for synchronisation and initialisation                                                                                                                            |         | 5000ms               |
| Analog-Register 150 (Alarm Counter) is being changed if new Alarm occurs or Alarm disappears. If Register-Value has changed since last request, read all Alarm-Informations at once. |         |                      |
| 2 Bits per Alarm                                                                                                                                                                     |         |                      |
| High-Bit                                                                                                                                                                             | Low-Bit |                      |
| 0                                                                                                                                                                                    | 0       | No Alarm             |
| 0                                                                                                                                                                                    | 1       | Acknowledged Alarm   |
| 1                                                                                                                                                                                    | 0       | Spare/not used       |
| 1                                                                                                                                                                                    | 1       | Unacknowledged Alarm |

| Register | Denomination                  | Alarm Priority | Alarm Description                                                                                                                                                                                                                             |
|----------|-------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1002     | LOLO Engine Speed             | Yellow (3)     | Engine is being stalled. The engine speed of the normally operating engine dropped below the limit from parameter 2.2500.027 Limit Engine Speed Low without any stop request. For safety reason the engine is stopped when this event occurs. |
| 1004     | HIHI Engine Overspeed (EIM)   | Red (2)        | Highest Limit of engine overspeed                                                                                                                                                                                                             |
| 1006     | HIHI Engine Overspeed SaSy    | Red (2)        | An engine overspeed was detected by the safety system. This alarm will normally cause an automatic emergency stop.                                                                                                                            |
| 1008     | SD Engine Speed Camshaft      | Yellow (4)     | Sensor defect on camshaft. ==> short circuit or cable breakage ==> Check sensor and cable (B1), if necessary replace it. After Engine restart follows curing of failure.                                                                      |
| 1010     | SD Engine Speed Crankshaft    | Yellow (4)     | Sensor defect on crankshaft. ==> short circuit or cable breakage ==> Check sensor and cable (B13), if necessary replace it. After Engine restart follows curing of failure.                                                                   |
| 1018     | SD ETC 1 Speed                | Yellow (4)     | Speed-sensor of basic charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.1), if necessary replace it.                                                                                                        |
| 1020     | HIHI ETC 1 Overspeed          | Red (2)        | Speed of basic charger too high (limit 2).                                                                                                                                                                                                    |
| 1022     | HI ETC 1 Speed                | Yellow (3)     | Speed of basic charger too high (limit 1).                                                                                                                                                                                                    |
| 1024     | SD ETC 2 Speed                | Yellow (4)     | Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.2), if necessary replace it.                                                                                                    |
| 1026     | HIHI ETC 2 Overspeed          | Red (2)        | Speed of 1st switchable charger too high (limit 2).                                                                                                                                                                                           |
| 1028     | HI ETC 2 Speed                | Yellow (3)     | Speed of 1st switchable charger too high (limit 1).                                                                                                                                                                                           |
| 1030     | SD ETC 3 Speed                | Yellow (4)     | Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.2), if necessary replace it.                                                                                                    |
| 1032     | HIHI ETC 3 Overspeed          | Red (2)        | Speed of 2nd switchable charger too high (limit 2).                                                                                                                                                                                           |
| 1034     | HI ETC 3 Speed                | Yellow (3)     | Speed of 2nd switchable charger too high (limit 1).                                                                                                                                                                                           |
| 1036     | SD ETC 4 Speed                | Yellow (4)     | Speed-sensor of switching charger defect. ==> short circuit or cable breakage ==> Check sensor and cable (B44.4), if necessary replace it.                                                                                                    |
| 1038     | HIHI ETC 4 Overspeed          | Red (2)        | Speed of 3rd switchable charger too high (limit 2).                                                                                                                                                                                           |
| 1040     | HI ETC 4 Speed                | Yellow (3)     | Speed of 3rd switchable charger too high (limit 1).                                                                                                                                                                                           |
| 1048     | HI ETC Speed Difference       | Yellow (4)     | Speed deviation between basic turbo charger and one of the switchable chargers.                                                                                                                                                               |
| 1110     | LO Starter Supply Voltage     | Yellow (4)     | The starter supply voltage is below the minimum limit. The starter supply voltage is not monitored during the start procedure.                                                                                                                |
| 1112     | SD ECU Power Supply Voltage   | Yellow (4)     | Internal ECU error. ==> electronic defect.                                                                                                                                                                                                    |
| 1114     | HIHI ECU Power Supply Voltage | Red (2)        | Power supply voltage too high (limit 2) ==> Check battery / generator                                                                                                                                                                         |
| 1116     | HI ECU Power Supply Voltage   | Yellow (4)     | Power supply voltage too high (limit 1) ==> Check battery / generator                                                                                                                                                                         |
| 1118     | LO ECU Power Supply Voltage   | Yellow (4)     | Power supply voltage too low (limit 1) ==> Check battery / generator                                                                                                                                                                          |
| 1120     | LOLO ECU Power Supply Voltage | Red (2)        | Power supply voltage too low (limit 2) ==> Check battery / generator                                                                                                                                                                          |
| 1122     | SD U-Power Driver Unit        | Yellow (4)     | Sensor defect of Injector Power driver unit. ==> Internal error of ECU7. Change ECU7.                                                                                                                                                         |
| 1124     | HIHI U-Power Driver Unit      | Yellow (3)     | Power driver voltage (injectors) too high (limit 2).                                                                                                                                                                                          |
| 1126     | HI U-Power Driver Unit        | Yellow (3)     | Power driver voltage (injectors) too high (limit 1).                                                                                                                                                                                          |
| 1128     | LO U-Power Driver Unit        | Yellow (4)     | Power driver voltage (injectors) too low (limit 1).                                                                                                                                                                                           |
| 1130     | LOLO U-Power Driver Unit      | Red (2)        | Power driver voltage (injectors) too low (limit 2).                                                                                                                                                                                           |
| 1132     | HI U-SPU Main Power Supply    | Yellow (3)     | The voltage of the SPU main power supply is too high.                                                                                                                                                                                         |
| 1134     | LO U-SPU Main Power Supply    | Yellow (3)     | The voltage of the SPU main power supply is too low.                                                                                                                                                                                          |



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|                                                                |                                                      |                                                 |
|----------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------|
| Änderungsbeschreibung/Description of Revision<br><b>war kZ</b> |                                                      | Kommt vor/Frequency                             |
| Buchst./Rev.Ltr.<br><b>-14</b>                                 | Änderungs-Nr./Revision Notice No.<br><b>PR047866</b> | Bearbeitungsstatus/Lifecycle<br><b>Released</b> |

| Register | Denomination                    | Alarm Priority | Alarm Description                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|---------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1136     | HI U-SPU Emerg.Power Supply     | Yellow (3)     | The voltage of the SPU emergency power supply is too high.                                                                                                                                                                                                                                                                                                                                                 |
| 1138     | LO U-SPU Emerg.Power Supply     | Yellow (3)     | The voltage of the SPU emergency power supply is too low.                                                                                                                                                                                                                                                                                                                                                  |
| 1140     | LO U-SPU Real Time Clock        | Yellow (3)     | The voltage of backup battery for time and date in the LOP is too low, logging features may have an incorrect time stamp. Replace the battery located in the SPU/LOP!                                                                                                                                                                                                                                      |
| 1142     | AL EMU 11V                      | Yellow (4)     | The internal 11V power supply of the engine monitoring extension unit (EMU) is out of range. This could cause improper readings of measured data and malfunctioning safety functions.                                                                                                                                                                                                                      |
| 1144     | AL EMU 7V                       | Yellow (4)     | The internal -7V power supply of the engine monitoring extension unit (EMU) is out of range. This could cause improper readings of measured data and malfunctioning safety functions.                                                                                                                                                                                                                      |
| 1146     | AL U-SPU Main Power Supply/Fail | Yellow (4)     | The main power supply of the LOP/SPU failed. The system is still full operative and is feed through the emergency power supply.                                                                                                                                                                                                                                                                            |
| 1148     | AL U-SPU Emerg. Pow Sup Fail    | Yellow (4)     | The emergency power supply of the LOP/SPU failed. The system is still full operative and is feed through the emergency power supply.                                                                                                                                                                                                                                                                       |
| 1150     | SD P-Lube Oil                   | Yellow (4)     | Lube oil pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5), if necessary replace it.                                                                                                                                                                                                                                                                             |
| 1152     | LO P-Lube Oil                   | Yellow (3)     | Pressure of lube oil too low (limit 1).                                                                                                                                                                                                                                                                                                                                                                    |
| 1154     | LOLO P-Lube Oil                 | Red (2)        | Pressure of lube oil too low (limit 2).                                                                                                                                                                                                                                                                                                                                                                    |
| 1156     | SD P-Lube Oil SaSy              | Yellow (4)     | The safety system sensor signal for lube oil pressure is outside the defined sensor signal range. The safety system lube oil pressure information is not available anymore for indication or monitoring. The safety functions related to this signal are not available anymore.                                                                                                                            |
| 1158     | LO P-Lube Oil SaSy              | Yellow (3)     | A low engine lube oil pressure was detected by the safety system. A worsening situation could trigger a safety reaction.                                                                                                                                                                                                                                                                                   |
| 1160     | LOLO P-Lube Oil SaSy            | Red (2)        | A very low engine lube oil pressure was detected by the safety system. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with an separate alarm.                                                                                                                                                                                                               |
| 1162     | TD P-Lube Oil                   | Yellow (4)     | The pressure difference between the redundant lube oil pressure sensors is too high.                                                                                                                                                                                                                                                                                                                       |
| 1164     | SD P-Lube Oil A-Side ETCs       | Yellow (4)     | Lube oil pressure-sensor of ETC on engine A-Side is defect. ==> short circuit or cable breakage ==> Check sensor and cable if necessary replace it.                                                                                                                                                                                                                                                        |
| 1166     | LO P-Lube Oil A-Side ETCs       | Yellow (3)     | A low lube oil pressure of the engine turbo charger on the A-side was detected. A worsening situation could trigger a safety reaction. This is the warning level.                                                                                                                                                                                                                                          |
| 1168     | LOLO P-Lube Oil A-Side ETCs     | Red (2)        | A very low lube oil pressure of the engine turbo charger on the A-side was detected. A worsening situation could trigger a safety reaction. This is the warning level. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with a separate alarm.                                                                                                                |
| 1170     | SD P-Lube Oil B-Side ETCs       | Yellow (4)     | Lube oil pressure-sensor of ETC on engine B-side is defect. ==> short circuit or cable breakage ==> Check sensor and cable if necessary replace it.                                                                                                                                                                                                                                                        |
| 1172     | LO P-Lube Oil B-Side ETCs       | Yellow (3)     | A low lube oil pressure of the engine turbo charger on the B-side was detected. A worsening situation could trigger a safety reaction. This is the warning level.                                                                                                                                                                                                                                          |
| 1174     | LOLO P-Lube Oil B-Side ETCs     | Red (2)        | A very low lube oil pressure of the engine turbo charger on the B-side was detected. A worsening situation could trigger a safety reaction. This is the warning level. This alarm will normally cause an automatic security shutdown. The safety reaction is alarmed with an separate alarm.                                                                                                               |
| 1176     | SD P-Oil after Level Pump       | Yellow (4)     | Sensor defect of lube oil pressure after level pump                                                                                                                                                                                                                                                                                                                                                        |
| 1178     | LO P-Oil after Level Pump       | Yellow (3)     | Low Limit of lube oil pressure after level pump                                                                                                                                                                                                                                                                                                                                                            |
| 1180     | SD P-Oil bef.High Press.Pump A  | Yellow (4)     | Sensor defect of lube oil pressure before high pressure pump in A                                                                                                                                                                                                                                                                                                                                          |
| 1182     | LO P-Oil bef.High Press.Pump A  | Yellow (3)     | Low Limit of lube oil pressure before high pressure pump A                                                                                                                                                                                                                                                                                                                                                 |
| 1184     | SD P-Oil bef.High Press.Pump B  | Yellow (4)     | Sensor defect of lube oil pressure before high pressure pump in B                                                                                                                                                                                                                                                                                                                                          |
| 1186     | LO P-Oil bef.High Press.Pump B  | Yellow (3)     | Low Limit of lube oil pressure before high pressure pump B                                                                                                                                                                                                                                                                                                                                                 |
| 1204     | SD P-Lube Oil before Filter     | Yellow (4)     | Pressure sensor for lube oil before filter defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5.3), if necessary replace it.                                                                                                                                                                                                                                                         |
| 1206     | SD P-Oil Filter Difference      | Yellow (4)     | Sensor for differential pressure of lube oil defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5), if necessary replace it.                                                                                                                                                                                                                                                         |
| 1208     | HIHI P-Oil Filter Difference    | Red (2)        | Differential pressure of oil filter too high (limit 2).                                                                                                                                                                                                                                                                                                                                                    |
| 1210     | HI P-Oil Filter Difference      | Yellow (3)     | Differential pressure of oil filter too high (limit 1).                                                                                                                                                                                                                                                                                                                                                    |
| 1226     | SD P-Oil Refill Pump            | Yellow (4)     | The pressure sensor signal for the oil refill pump pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal are not available anymore.                                                                                                                             |
| 1228     | LO P-Oil Refill Pump            | Yellow (3)     | The oil pressure after the oil refill pump which is pumping oil out and back into the oil pan was low for a certain period (several minutes). As long as the pump is able to pump oil out and back into the oil pan, there is enough oil in the engine. If the oil level is below the end of the "pump snorkel" the pump will not produce pressure within a certain time and this alarm will be triggered. |
| 1232     | SD P-Coolant                    | Yellow (4)     | Coolant pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B16), if necessary replace it.                                                                                                                                                                                                                                                                             |
| 1234     | LO P-Coolant                    | Yellow (3)     | Coolant pressure too low (limit 1) ==> check cooling cycle.                                                                                                                                                                                                                                                                                                                                                |
| 1236     | LOLO P-Coolant                  | Red (2)        | Coolant pressure too low (limit 2) ==> Engine stop or limitation of the injection quantity ==> check cooling cycle                                                                                                                                                                                                                                                                                         |
| 1242     | SD P-Raw Water                  | Yellow (4)     | The pressure sensor signal for the raw water pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal are not available anymore.                                                                                                                                   |
| 1244     | LO P-Raw Water                  | Yellow (3)     | The raw water pressure after the pump is too low                                                                                                                                                                                                                                                                                                                                                           |
| 1246     | SD P-Fuel after Filter          | Yellow (4)     | Fuel pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B34), if necessary replace it.                                                                                                                                                                                                                                                                                |
| 1248     | LO P-Fuel                       | Yellow (3)     | Fuel supply pressure too low (limit 1) ==> Check filter, fuel on low-pressure side                                                                                                                                                                                                                                                                                                                         |

|                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                  |                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------|-------------------------------------------------|
| <br><b>MTU Friedrichshafen GmbH</b>                                                                                                                                                                                                                                                                                             |  | Zeichnungs-Nr./Drawing No.<br><b>ZNG00019138</b> | Blatt/ Sheet<br><b>7</b><br>von/of<br><b>27</b> |
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|                                                         |                                               |                                          |
|---------------------------------------------------------|-----------------------------------------------|------------------------------------------|
| Änderungsbeschreibung/Description of Revision<br>war kZ |                                               | Kommt vor/Frequency                      |
| Buchst./Rev.Ltr.<br>-14                                 | Änderungs-Nr./Revision Notice No.<br>PR047866 | Bearbeitungsstatus/Lifecycle<br>Released |

| Register | Denomination                   | Alarm Priority | Alarm Description                                                                                                                                                                                                                                                                                                           |
|----------|--------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1250     | LOLO P-Fuel                    | Red (2)        | Fuel supply pressure too low (limit 2) ==> Check filter (low-pressure side)                                                                                                                                                                                                                                                 |
| 1252     | SD P-Fuel before Filter        | Yellow (4)     | Fuel pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B5.3), if necessary replace it.                                                                                                                                                                                                |
| 1254     | SD Fuel Prefilter 1            | Yellow (4)     | The self test function of the fuel prefilter sensor 1 detected that the sensor does not respond as specified. The sensor might be defect or the wiring is defective. The SD alarm is reset if a successful self test was carried out. A self test is typically performed every 15min (the exact time might vary).           |
| 1256     | SD Fuel Prefilter 2            | Yellow (4)     | The self test function of the fuel prefilter sensor 2 detected that the sensor does not respond as specified. The sensor might be defect or the wiring is defective. The SD alarm is reset if a successful self test was carried out. A self test is typically performed every 15min (the exact time might vary).           |
| 1258     | HI P-Fuel Int. Filter Diff.    | Yellow (3)     | Fuel Prefilter Difference to high                                                                                                                                                                                                                                                                                           |
| 1260     | SD P-Fuel Filter Difference    | Yellow (4)     | Sensor for differential pressure of fuel defect. ==> only if SD-Alarms Fuel b. Filter or Fuel after Filter                                                                                                                                                                                                                  |
| 1262     | HI P-Fuel Filter Difference    | Yellow (3)     | Differential pressure of fuel filter too high (limit 1).                                                                                                                                                                                                                                                                    |
| 1264     | SD P-Fuel bef. Low Press. Pump | Yellow (4)     | Sensor defect of fuel pressure before low pressure pump                                                                                                                                                                                                                                                                     |
| 1266     | LO P-Fuel bef. Low Press. Pump | Yellow (3)     | Low Limit of fuel pressure before low pressure pump                                                                                                                                                                                                                                                                         |
| 1268     | SD P-Fuel Common Rail          | Yellow (4)     | Rail pressure-sensor defect. ==> High-pressure controller emergency operation ==> short circuit or cable breakage ==> Check sensor and cable (B48), if necessary replace it.                                                                                                                                                |
| 1270     | HI P-Fuel Common Rail          | Yellow (3)     | Rail pressure > set point value => DBR reduction, shift of start of injection delayed (==> interphase transformer sticks or connections of the interphase transformer)                                                                                                                                                      |
| 1272     | LOLO P-Fuel Common Rail        | Red (2)        | Rail pressure < setpoint value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)                                                                                                                                                                                                   |
| 1274     | LO P-Fuel Common Rail          | Yellow (3)     | Rail pressure < set point value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)                                                                                                                                                                                                  |
| 1276     | SD P-Fuel Common Rail B        | Yellow (4)     | Rail pressure-sensor defect. ==> High-pressure controller emergency operation ==> short circuit or cable breakage ==> Check sensor and cable (B48), if necessary replace it.                                                                                                                                                |
| 1278     | HI P-Fuel Common Rail B        | Yellow (3)     | High Limit of fuel pressure in rail B                                                                                                                                                                                                                                                                                       |
| 1280     | LOLO P-Fuel Common Rail B      | Red (2)        | Rail pressure on Rail B < setpoint value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)                                                                                                                                                                                         |
| 1282     | LO P-Fuel Common Rail B        | Yellow (3)     | Rail pressure < setpoint value => DBR reduction (==> interphase transformer defective or leakage in high-pressure system)                                                                                                                                                                                                   |
| 1296     | SD P-Fuel Return Path          | Yellow (4)     | Sensor defect of fuel pressure in return path                                                                                                                                                                                                                                                                               |
| 1298     | HI P-Fuel Return Path          | Yellow (3)     | High Limit of fuel pressure in return path                                                                                                                                                                                                                                                                                  |
| 1300     | SD P-Crankcase                 | Yellow (4)     | Crankcase pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B50), if necessary replace it.                                                                                                                                                                                            |
| 1302     | HIHI P-Crankcase               | Red (2)        | Crankcase pressure too high (limit 2).                                                                                                                                                                                                                                                                                      |
| 1304     | HI P-Crankcase                 | Yellow (3)     | Crankcase pressure too high (limit 1).                                                                                                                                                                                                                                                                                      |
| 1306     | LO P-Intake Air After FilterB  | Yellow (3)     | Low Limit of intake air pressure B reached                                                                                                                                                                                                                                                                                  |
| 1308     | SD P-Charge Air A              | Yellow (4)     | Charge air pressure-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B10), if necessary replace it.                                                                                                                                                                                           |
| 1310     | HIHI P-Charge Air A            | Red (2)        | Pressure of charge air too high (limit 2).                                                                                                                                                                                                                                                                                  |
| 1312     | HI P-Charge Air A              | Yellow (3)     | Pressure of charge air too high (limit 1).                                                                                                                                                                                                                                                                                  |
| 1314     | SD P-Charge Air B              | Yellow (4)     | Charge Air Pressur B-Side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B10.11), if necessary replace it. After Engine restart follows curing of failure.                                                                                                                                         |
| 1316     | HIHI P-Charge Air B            | Red (2)        | Highest Limit of charge air pressure in B reached                                                                                                                                                                                                                                                                           |
| 1318     | HI P-Charge Air B              | Yellow (3)     | High Limit of charge air pressure in B reached                                                                                                                                                                                                                                                                              |
| 1324     | SD P-Start Air                 | Yellow (4)     | The pressure sensor signal for the start air pressure monitoring is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured (safety) functions related to this signal (start interlock if start air pressure is too low) are not available anymore. |
| 1326     | LO P-Start Air                 | Yellow (4)     | The pressure for the air starter is too low.                                                                                                                                                                                                                                                                                |
| 1328     | LOLO P-Start Air               | Yellow (3)     | The pressure for the air starter is very low and a start is interlocked. The limit for that alarm should be set in a way that air for at least one start is remaining. This is to prevent emptying the air supply by (automatic) repeated starts. The start interlock can be disabled by activating "Override"              |
| 1386     | SD T-Lube Oil                  | Yellow (4)     | Lube oil temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B7), if necessary replace it.                                                                                                                                                                                           |
| 1388     | HIHI T-Lube Oil                | Red (2)        | Lube oil temperature to high (limit 2).                                                                                                                                                                                                                                                                                     |
| 1390     | HI T-Lube Oil                  | Yellow (3)     | Lube oil temperature too high (limit 1).                                                                                                                                                                                                                                                                                    |
| 1394     | SD T-Lube Oil ETCs             | Yellow (4)     | T-Lube Oil turbo charger sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable, if necessary replace it.                                                                                                                                                                                            |
| 1396     | HIHI T-Lube Oil ETCs           | Red (2)        | Lube oil temperature of the basic charger is too high (above second limit).                                                                                                                                                                                                                                                 |
| 1398     | HI T-Lube Oil ETCs             | Yellow (3)     | Lube oil temperature of the basic charger is too high (above first limit).                                                                                                                                                                                                                                                  |
| 1418     | SD T-Fuel bef.Low Press.Pump   | Yellow (4)     | T-Fuel before Engine sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable, if necessary replace it.                                                                                                                                                                                                |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------|-------------------------------------------------|
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| Register | Denomination                   | Alarm Priority | Alarm Description                                                                                                                                                                                                                                                                   |
|----------|--------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1420     | HIHI T-Fuel before LP Pump     | Red (2)        | Highest Limit of fuel temperature before low pressure pump reached                                                                                                                                                                                                                  |
| 1422     | HI T-Fuel before LP Pump       | Yellow (3)     | Fuel temperature before the low pressure fuel pump is too high.                                                                                                                                                                                                                     |
| 1424     | SD T-Fuel                      | Yellow (4)     | Fuel temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B33), if necessary replace it.                                                                                                                                                      |
| 1426     | HIHI T-Fuel                    | Red (2)        | Fuel temperature too high (limit 2).                                                                                                                                                                                                                                                |
| 1428     | HI T-Fuel                      | Yellow (3)     | Fuel temperature too high (limit 1)                                                                                                                                                                                                                                                 |
| 1430     | SD T-Fuel Common Rail B        | Yellow (4)     | Sensor defect of fuel temperature in Rail B                                                                                                                                                                                                                                         |
| 1432     | HIHI T-Fuel Common Rail B      | Red (2)        | Highest Limit of fuel temperature in rail B reached                                                                                                                                                                                                                                 |
| 1434     | HI T-Fuel Common Rail B        | Yellow (3)     | High Limit of fuel temperature in rail B reached                                                                                                                                                                                                                                    |
| 1436     | SD T-Coolant                   | Yellow (4)     | Coolant temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B6), if necessary replace it.                                                                                                                                                    |
| 1438     | HIHI T-Coolant                 | Red (2)        | Coolant temperature too high (limit 2) ==> Check circulation of coolant                                                                                                                                                                                                             |
| 1440     | HI T-Coolant                   | Yellow (3)     | Coolant temperature too high (limit 1) ==> Check circulation of coolant                                                                                                                                                                                                             |
| 1442     | SD T-Coolant SaSy              | Yellow (4)     | The safety system sensor signal for coolant temperature is outside the defined sensor signal range. The safety system coolant temperature information is not available anymore for indication or monitoring. The safety functions related to this signal are not available anymore. |
| 1444     | HIHI T-Coolant SaSy            | Red (2)        | A very high engine coolant temperature was detected by the safety system. This alarm will normally cause an automatic safety function. The safety reaction is alarmed with an separate alarm.                                                                                       |
| 1446     | HI T-Coolant SaSy              | Yellow (4)     | A high engine coolant temperature was detected by the safety system. A worsening situation could trigger a safety reaction.                                                                                                                                                         |
| 1448     | TD T-Coolant                   | Yellow (4)     | The temperature difference between the redundant coolant temperature sensors is too high.                                                                                                                                                                                           |
| 1450     | SD T-Raw Water after Pump      | Yellow (4)     | Sensor defect of raw water temperature after pump                                                                                                                                                                                                                                   |
| 1452     | HI T-Raw Water after Pump      | Yellow (3)     | High Limit of raw water temperature after pump reached                                                                                                                                                                                                                              |
| 1454     | SD T-Intake Air                | Yellow (4)     | Intake air temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B3), if necessary replace it.                                                                                                                                                 |
| 1456     | HIHI T-Intake Air              | Red (2)        | Highest Limit of intake air temperature reached                                                                                                                                                                                                                                     |
| 1458     | HI T-Intake Air                | Yellow (3)     | Intake Air temperature too high (limit 1).                                                                                                                                                                                                                                          |
| 1460     | SD P-Intake Air After FilterB  | Yellow (4)     | Intake air pressure sensor and cable , if necessary replace it.                                                                                                                                                                                                                     |
| 1464     | SD T-Charge Air                | Yellow (4)     | Charge air temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B9), if necessary replace it.                                                                                                                                                 |
| 1466     | HIHI T-Charge Air              | Red (2)        | Charge air temperature too high (limit 2) ==> Check Intercooler                                                                                                                                                                                                                     |
| 1468     | HI T-Charge Air                | Yellow (3)     | Charge air temperature too high (limit 1) ==> Check Intercooler                                                                                                                                                                                                                     |
| 1470     | SD T-Charge Air B              | Yellow (4)     | Charge Air B temperature-sensor defect. ==> short circuit or cable breakage ==> Check sensor and cable (B3), if necessary replace it.                                                                                                                                               |
| 1472     | HIHI T-Charge Air B            | Red (2)        | Highest Limit of charge air temperature in B reached                                                                                                                                                                                                                                |
| 1474     | HI T-Charge Air B              | Yellow (3)     | High Limit of charge air temperature in B reached                                                                                                                                                                                                                                   |
| 1476     | SD T-Charge Air Seq Ctrl Valve | Yellow (4)     | Temperature-sensor of recirculated charge air defect. ==> short circuit or cable breakage ==> Check sensor and cable (B49), if necessary replace it.                                                                                                                                |
| 1478     | HIHI T-Charge Air Seq Ctrl Val | Red (2)        | Umblasen' temperature too high (limit 2).                                                                                                                                                                                                                                           |
| 1480     | HI T-Charge Air Seq Ctrl Valve | Yellow (3)     | Umblasen' temperature too high (limit 1).                                                                                                                                                                                                                                           |
| 1482     | SD T-Exhaust Combined A        | Yellow (4)     | Exhaust gas temperature-sensor on A-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.21), if necessary replace it.                                                                                                                                   |
| 1484     | HIHI T-Exhaust Combined A      | Red (2)        | Exhaust gas temperature (A-side) too high (limit 2).                                                                                                                                                                                                                                |
| 1486     | HI T-Exhaust Combined A        | Yellow (3)     | Exhaust gas temperature (A-side) too high (limit 1).                                                                                                                                                                                                                                |
| 1488     | SD T-Exhaust Combined B        | Yellow (4)     | Exhaust gas temperature-sensor on B-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.22), if necessary replace it.                                                                                                                                   |
| 1490     | HIHI T-Exhaust Combined B      | Red (2)        | Exhaust gas temperature (B-side) too high (limit 2).                                                                                                                                                                                                                                |
| 1492     | HI T-Exhaust Combined B        | Yellow (3)     | Exhaust gas temperature (B-side) too high (limit 1).                                                                                                                                                                                                                                |
| 1494     | SD T-Exhaust Combined C        | Yellow (4)     | Exhaust gas temperature-sensor on C-side defect. ==> short circuit or cable breakage ==> Check sensor and cable (B4.23), if necessary replace it. After Engine restart follows curing of failure.                                                                                   |
| 1496     | HIHI T-Exhaust Combined C      | Red (2)        | Exhaust gas temperature (C-side) too high (limit 2).                                                                                                                                                                                                                                |
| 1498     | HI T-Exhaust Combined C        | Yellow (3)     | Exhaust gas temperature (C-side) too high (limit 1).                                                                                                                                                                                                                                |



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| Register | Denomination              | Alarm Priority | Alarm Description                                                                                                                                                                                                                                                                    |
|----------|---------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1500     | SD T-Exhaust Combined D   | Yellow (4)     | Exhaust gas temperature-sensor on D-side defect. ==> short circuit or cable breakage<br>==> Check sensor and cable (B4.24), if necessary replace it. After Engine restart follows curing of failure.                                                                                 |
| 1502     | HIHI T-Exhaust Combined D | Red (2)        | Exhaust gas temperature (D-side) too high (limit 2).                                                                                                                                                                                                                                 |
| 1504     | HI T-Exhaust Combined D   | Yellow (3)     | Exhaust gas temperature (D-side) too high (limit 1).                                                                                                                                                                                                                                 |
| 1506     | SD T-Exhaust A1           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1508     | HI T-Exhaust A1           | Yellow (3)     | The temperature of cylinder A1 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1510     | LO T-Exhaust A1           | Yellow (3)     | The temperature of cylinder A1 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1512     | SD T-Exhaust A2           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1514     | HI T-Exhaust A2           | Yellow (3)     | The temperature of cylinder A2 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1516     | LO T-Exhaust A2           | Yellow (3)     | The temperature of cylinder A2 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1518     | SD T-Exhaust A3           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1520     | HI T-Exhaust A3           | Yellow (3)     | The temperature of cylinder A3 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1522     | LO T-Exhaust A3           | Yellow (3)     | The temperature of cylinder A3 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1524     | SD T-Exhaust A4           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1526     | HI T-Exhaust A4           | Yellow (3)     | The temperature of cylinder A4 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1528     | LO T-Exhaust A4           | Yellow (3)     | The temperature of cylinder A4 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1530     | SD T-Exhaust A5           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1532     | HI T-Exhaust A5           | Yellow (3)     | The temperature of cylinder A5 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1534     | LO T-Exhaust A5           | Yellow (3)     | The temperature of cylinder A5 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1536     | SD T-Exhaust A6           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1538     | HI T-Exhaust A6           | Yellow (3)     | The temperature of cylinder A6 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1540     | LO T-Exhaust A6           | Yellow (3)     | The temperature of cylinder A6 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1542     | SD T-Exhaust A7           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1544     | HI T-Exhaust A7           | Yellow (3)     | The temperature of cylinder A7 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1546     | LO T-Exhaust A7           | Yellow (3)     | The temperature of cylinder A7 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1548     | SD T-Exhaust A8           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1550     | HI T-Exhaust A8           | Yellow (3)     | The temperature of cylinder A8 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1552     | LO T-Exhaust A8           | Yellow (3)     | The temperature of cylinder A8 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1554     | SD T-Exhaust A9           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1556     | HI T-Exhaust A9           | Yellow (3)     | The temperature of cylinder A9 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1558     | LO T-Exhaust A9           | Yellow (3)     | The temperature of cylinder A9 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1560     | SD T-Exhaust A10          | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1562     | HI T-Exhaust A10          | Yellow (3)     | The temperature of cylinder A10 compared to the average exhaust temperature is too high.                                                                                                                                                                                             |
| 1564     | LO T-Exhaust A10          | Yellow (3)     | The temperature of cylinder A10 compared to the average exhaust temperature is too low.                                                                                                                                                                                              |
| 1566     | SD T-Exhaust B1           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |
| 1568     | HI T-Exhaust B1           | Yellow (3)     | The temperature of cylinder B1 compared to the average exhaust temperature is too high.                                                                                                                                                                                              |
| 1570     | LO T-Exhaust B1           | Yellow (3)     | The temperature of cylinder B1 compared to the average exhaust temperature is too low.                                                                                                                                                                                               |
| 1572     | SD T-Exhaust B2           | Yellow (4)     | The sensor signal for the single exhaust temperature of the specified cylinder is outside the defined sensor signal range. This measuring point is not available anymore for indication or monitoring. Configured safety functions related to this signal are not available anymore. |



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